

# HDOC Day-35

Write a menu-driven C program to read a variable-length square integer array and perform the following operations using separate functions:

1. Determine whether all the elements on its main diagonal are distinct.
2. Find the sum of its main diagonal elements.
3. Display its elements using a diagonal traversal. Begin with the element in the top-left corner and traverse successive diagonals alternately in the upward-right and downward-left directions.
4. Exit.

**Example:** For the following input matrix

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

1. Main diagonal elements: 1 5 9. All the main diagonal elements are distinct.
2. Sum of main diagonal elements: 15
3. Diagonal traversal: 1 2 4 7 5 3 6 8 9

## 1. Algorithm

### Step 1: Start

### Step 2:

Read the size n of the square matrix.

### Step 3:

Read the  $n \times n$  matrix elements.

### Step 4:

Display the menu:

- Choice 1 → Check whether main diagonal elements are distinct.
- Choice 2 → Find the sum of main diagonal elements.
- Choice 3 → Perform diagonal traversal.
- Choice 4 → Exit.

**Step 5:**

Read the user's choice.

**Step 6: If choice = 1**

- Compare every main diagonal element with the other diagonal elements.
- If any two are equal, display "**Main diagonal elements are not distinct.**"
- Otherwise, display "**All the main diagonal elements are distinct.**"

**Step 7: If choice = 2**

- Initialize sum = 0.
- Add  $a[i][i]$  for every  $i$ .
- Display the sum.

**Step 8: If choice = 3**

- Traverse the matrix diagonal by diagonal.
- Traverse alternate diagonals in:
  - Upward-right direction.
  - Downward-left direction.
- Display the elements.

**Step 9: If choice = 4**

Display "**Exiting the program...**"

**Step 10:**

For any other choice, display "**Invalid choice.**"

**Step 11:**

Repeat the menu until the user chooses 4.

**Step 12: Stop.****Test Case Table:**

| Test Case | Matrix / Input        | Choice | Expected Output                                 |
|-----------|-----------------------|--------|-------------------------------------------------|
| 1         | 1 2 3 / 4 5 6 / 7 8 9 | 1      | Main diagonal: 1 5 9; All elements are distinct |
| 2         | 1 2 3 / 4 5 6 / 7 8 9 | 2      | Sum = 15                                        |

|   |                       |   |                                               |
|---|-----------------------|---|-----------------------------------------------|
| 3 | 1 2 3 / 4 5 6 / 7 8 9 | 3 | Traversal= 1 2 4 7 3<br>6 8 9                 |
| 4 | 1 2 3 / 4 5 5 / 7 8 9 | 1 | Main diagonal<br>elements are not<br>distinct |
| 5 | 5 2 3 / 4 5 6 / 7 8 5 | 1 | Main diagonal<br>elements are not<br>distinct |
| 6 | 1 2 / 3 4             | 3 | Traversal = 1 2 3 4                           |
| 7 | Any valid matrix      | 4 | Exiting the program                           |
| 8 | Any valid matrix      | 5 | Invalid choice                                |

Compilation and Execution:

The following program has been compiled and executed successfully.

Test Case 1:

```

(kali㉿kali)-[~]
└─$ ./a.out
Enter the size of square matrix: 3

Enter the elements of the matrix:
1 2 3 4 5 6 7 8 9

===== MENU =====
1. Check whether main diagonal elements are distinct
2. Find sum of main diagonal elements
3. Display diagonal traversal
4. Exit
=====

Enter your choice: 1

Main diagonal elements: 1 5 9
All the main diagonal elements are distinct.

```

Test Case 2:

```
===== MENU =====
1. Check whether main diagonal elements are distinct
2. Find sum of main diagonal elements
3. Display diagonal traversal
4. Exit
=====
Enter your choice: 2

Sum of main diagonal elements = 15
```

Test Case 3:

```
===== MENU =====
1. Check whether main diagonal elements are distinct
2. Find sum of main diagonal elements
3. Display diagonal traversal
4. Exit
=====
Enter your choice: 3

Diagonal Traversal: 1 2 4 7 5 3 6 8 9
```

Test Case 4:

```
(kali㉿kali)-[~]
└─$ ./a.out
Enter the size of square matrix: 3

Enter the elements of the matrix:
1 2 3 4 5 5 7 8 5

===== MENU =====
1. Check whether main diagonal elements are distinct
2. Find sum of main diagonal elements
3. Display diagonal traversal
4. Exit
=====
Enter your choice: 1

Main diagonal elements: 1 5 5
Main diagonal elements are not distinct.
```

Test Case 5:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter the size of square matrix: 3  
  
Enter the elements of the matrix:  
5 2 3 4 5 6 7 8 5  
  
===== MENU =====  
1. Check whether main diagonal elements are distinct  
2. Find sum of main diagonal elements  
3. Display diagonal traversal  
4. Exit  
===== MENU =====  
Enter your choice: 1  
  
Main diagonal elements: 5 5 5  
Main diagonal elements are not distinct.
```

Test Case 6:

```
(kali㉿kali)-[~]  
$ ./a.out  
Enter the size of square matrix: 2  
  
Enter the elements of the matrix:  
1 2 3 4  
  
===== MENU =====  
1. Check whether main diagonal elements are distinct  
2. Find sum of main diagonal elements  
3. Display diagonal traversal  
4. Exit  
===== MENU =====  
Enter your choice: 3  
  
Diagonal Traversal: 1 2 3 4
```

Test Case 7:

```
===== MENU =====
1. Check whether main diagonal elements are distinct
2. Find sum of main diagonal elements
3. Display diagonal traversal
4. Exit
=====
Enter your choice: 4

Exiting the program...
```

Test Case 8:

```
(kali㉿kali)-[~]
└─$ ./a.out
Enter the size of square matrix: 3

Enter the elements of the matrix:
1 2 3 4 5 6 7 8 9

===== MENU =====
1. Check whether main diagonal elements are distinct
2. Find sum of main diagonal elements
3. Display diagonal traversal
4. Exit
=====
Enter your choice: 5

Invalid choice! Please enter 1-4.
```